FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2024

TEST DATE: 31-03-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. $0 \le x \le \frac{R}{2}$, $i_1 = \int_{x=0}^{R/2} JdA$, $i_2 = \int_{R/2}^{R} JdA$

$i = i_1 + i_2$

2. C

Sol. $F = -\frac{dU}{dr}$

$F = kr = \frac{mv^2}{r}$

$v^2 = \frac{k}{m}r^2$

$v = \sqrt{\frac{k}{m}}r$

Also, $mvr = \frac{nh}{2\pi}$

$m\sqrt{\frac{k}{m}}r^2 = \frac{nh}{2\pi}$

$r \propto \sqrt{n}$

Energy = P.E. + K.E.

$= \frac{1}{2}kr^2 + \frac{1}{2}m\frac{k}{m}r^2$

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Energy = kr^2
$E \propto n$

3. A

Sol. The process is isochoric,
Let t be the temperature at any instant 't'

$$\frac{dQ}{dt} = \frac{kA(2T_0 - T)}{d} = \frac{5}{2}\frac{RdT}{dt}$$

$$\int_{T_0}^{T} \frac{dT}{2T_0 - T} = \int_0^t \frac{2}{5}\frac{kA}{dR}dt$$

4. C

Sol. $a = \frac{Ae^{-2r}}{r^3}$

Let $X = e^{-2r}$

$\ell nX = -2r$

$$\left|\frac{\Delta X}{X}\right| = |-2\Delta r|$$

$$\frac{\Delta a}{a} = \frac{\Delta x}{x} + \frac{3\Delta r}{r}$$

$$\frac{\Delta a}{a} = 2\Delta r + \frac{3\Delta r}{r}$$

$$\frac{\Delta a}{a} = 2(0.05) + 3\frac{(0.05)}{1} = 0.1 + 0.15$$

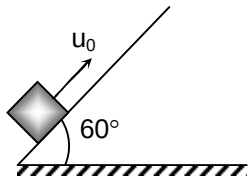
$$100 \times \frac{\Delta a}{a} = 0.25 \times 10 = 25\%$$

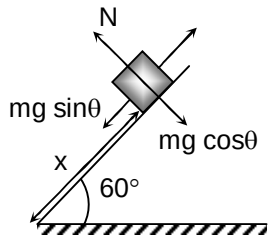
5. D

Sol. As half the length of candle always remain submerged in water therefore density of material of the candle is half the density of water. Then the Candle will come up at rate $2mmh^{-1.}$

6. C

Sol. Let x_0 be the distance moved by the box along the incline, when it comes to rest.





$W_f + W_g = \Delta E$

$$\Rightarrow \int dW = \int \left[\mu mg\cos\alpha + mg\sin\theta\right]dx$$

$$W = -\int_0^{x_0} (kxmg\cos\alpha + mg\sin\alpha)dx$$

$$W = -\frac{kx_0^2}{2}gm\cos\alpha - mg\sin\alpha x_0 = -\frac{1}{2}mu_0^2$$

$$\therefore\ u_0^2 = g\left[2\sin\alpha + kx_0\cos\alpha\right]x_0$$

at $x = x_0$, the body tends to slip downward so
$mg \sin \alpha = \mu mg \cos \alpha$
$\sin\alpha = kx_0\cos\alpha$
Hence minimum velocity v_0 is given by

$$u_0^2 = g[2\sin\alpha + \sin\alpha]\frac{\sin\alpha}{k\cos\alpha}$$

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$$= \frac{10(3)\sin^2 60°}{15(\cos 60°)}$$

$$\Rightarrow \quad u_0 = \sqrt{3} \text{ m/s}$$

7. B

Sol. ray is emerging from hemisphere at 0°.

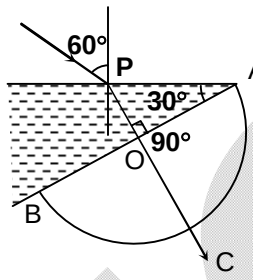
Snell's law at P

$\sin 60° = \sqrt{3}\sin r$

$r = 30°$

So, $\angle POA = 90°$

$PA = \frac{2R}{\sqrt{3}}$



8. C

Sol. $\text{Least count} = \frac{\text{Pitch}}{\text{number of divisions on circular scale}}$

$$= \frac{1\,\text{mm}}{100} = 0.01\,\text{mm} = 0.001\,\text{cm}$$

Zero error = – 0.03 mm

Measurement = 3.76 – (–0.03) = 3.79 mm

9. B

Sol. Acceleration of A is a $\sin\theta$ downwards

$mg - N = ma\sin\theta$

$N = mg - ma\sin\theta$

Velocity of A is v $\sin\theta$ downward

Power = $\vec{N}\cdot\vec{v} = -m(g - a\sin\theta)v\sin\theta$

10. C

Sol. deviation through first prism, $S_1 = (\mu_2 - 1)\theta$

deviation through second prism, $S_2 = -(\mu_1 - 1)\theta$

$S = S_1 + S_2 = (\mu_2 - \mu_1)\theta$

11. D

Sol. $\tan 60° = \frac{PQ}{CP}$

$\sqrt{3}CP = PQ = \frac{\sqrt{3}R}{\sqrt{2}}$

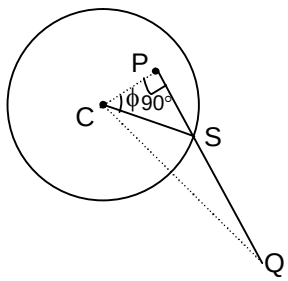
$CP = \frac{R}{\sqrt{2}}$

$CQ = \sqrt{\frac{3R^2}{2} + \frac{R^2}{2}}$

$CQ = \sqrt{2}R$

$\cos\phi = \frac{CP}{CS} = \frac{R/\sqrt{2}}{R}$

$\phi = 45°$



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$\therefore\ \theta = 15°,\ PS = \dfrac{R}{\sqrt{2}}$

Area of CPST = area of CPS + area of CST

$= \dfrac{1}{2}\left(\dfrac{R}{\sqrt{2}}\right)^2 + \dfrac{R^2}{2}\left[\dfrac{15°\pi}{180°}\right] = \dfrac{R^2}{24}(\pi + 6)m^2$

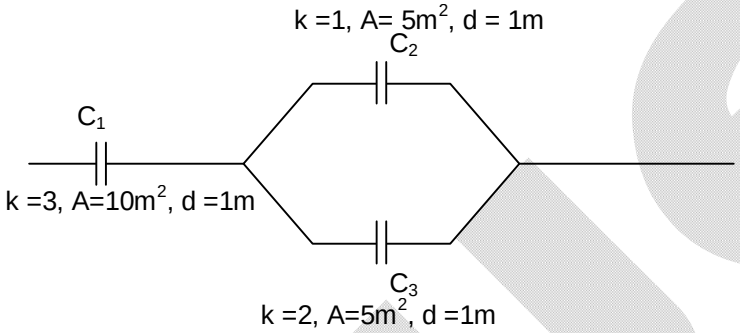
$\text{emf} = \dfrac{AdB}{dt} = \dfrac{R^2}{24}(\pi + 6)(4) = \dfrac{R^2}{6}(\pi + 6)$ volts

12. C

Sol. When polarity of Ge diode is reversed Ge diode is reverse biased, now for conduction potential difference across Si becomes 0.7V. So the potential difference across R_L = 12 – 0.7 = 11.3 V = V_y.

13. A

Sol.



14. D

Sol. $(2\mu_0 - \mu_0) + (-\mu_0 + 2\mu_0) + (-2\mu_0 + 4\mu_0) + (4\mu_0 - 2\mu_0) = \mu_0 I$

15. C

Sol. $50\cos 37° = \dfrac{v^2}{R}$

$40 = \dfrac{v^2}{10}$

16. A

Sol. $mgh = \int_0^d F dx$

17. A

Sol. Use work energy theorem till string become taut.

Work done by qE = change in kinetic energy

$qE\ell = \dfrac{1}{2}mv^2 - 0$

$v = \sqrt{\dfrac{2qE\ell}{m}}$

When string becomes taut an impulse is received and speed of the bob becomes

$v\cos 30° = \sqrt{\dfrac{3qE\ell}{2m}}$

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Further using work energy theorem speed of the bob becomes $v' = \sqrt{\frac{5qE\ell}{2m}}$

$\therefore$ time period = $\frac{2\pi\ell}{v'}$

After first half revolution in magnetic field work done by qE = change in kinetic energy

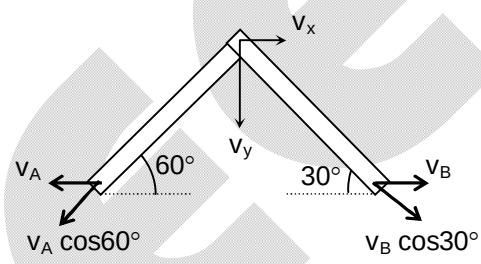
$qE(2\ell) = \frac{mv_1^2}{2} - \frac{m}{2}\left(\frac{5qE\ell}{m}\right)$

$\therefore$ angular speed of 1st revolution = $\sqrt{\frac{5qE}{2m\ell}}$

18. C

Sol. $v_x \cos 30° + v_y \cos 60° = v_B \cos 30°$

$V_y \cos 30° - v_x \cos 60° = v_A \cos 60°$



19. A

Sol. centre of mass should be at the same height

20. C

Sol. for PV^x = constant

$C = \frac{R}{\gamma - 1} - \frac{R}{x - 1}$

At C = 0 at x′

$C = 0 \Rightarrow x' = \gamma = 5/3$

At x = 0, C = C″

$C'' = \frac{R}{\frac{5}{3} - 1} - \frac{R}{-1} = \frac{5R}{2}$

SECTION – B

21. 4

Sol. volume of cylinder remains constant

$\frac{P_0}{T_0} = \frac{P}{2T_0}$

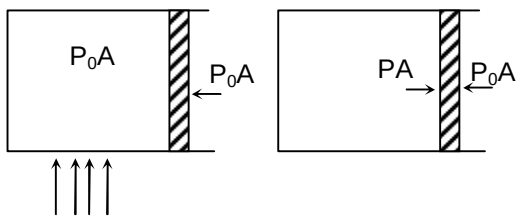
$\Rightarrow$ **$P = 2P_0$**

and $PA = P_0A + f$

$f = (P - P_0)A$

$(P - P_0)A = \int \mu\left(\frac{dN}{d\ell}\right)d\ell = \frac{dN}{d\ell}\mu(2\pi r)$

$\frac{dN}{d\ell} = \frac{(P - P_0)A}{\mu(2\pi r)}$



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22. 2

Sol. Due to refraction at the interface of glass slab and water image will be formed at

$$\frac{4/3}{v}-\frac{3/2}{-45}=0$$

$v = -40$ cm

for spherical mirror $u = -60$ cm

$$\frac{1}{v'}-\frac{1}{60}=-\frac{1}{20}$$

$\Rightarrow v' = -30$ cm

Again there will be refraction at water glass surface

$$\frac{3/2}{v}-\frac{4/3}{10}=0 \Rightarrow v=\frac{45}{4}$$

Again refraction at glass air interface $\frac{1}{v}-\frac{-(3/2)}{-(135/4)}=0$

$v = -22.5$ cm $= \frac{-45}{x}$

$\therefore x = 2$

23. 2

Sol. Velocity while the block climbs for the second time

$v' = v\cos^4 60°$ (an inelastic collision)

$$\frac{1}{2}m(v')^2 = mgh'$$

$$h' = \frac{v^2}{2g}\cos^8(60°) = h\cos^8 60° = \frac{512}{2^8} = 2m$$

24. 3

Sol. For the rod

Point A : $v_x + \frac{\omega R}{\sqrt{2}} = \frac{3v}{\sqrt{2}}$

Point B: $v_x - \frac{\omega R}{\sqrt{2}} = \frac{v}{\sqrt{2}}$

On solving $\omega = v/R$

and $v_x = \sqrt{2}v$

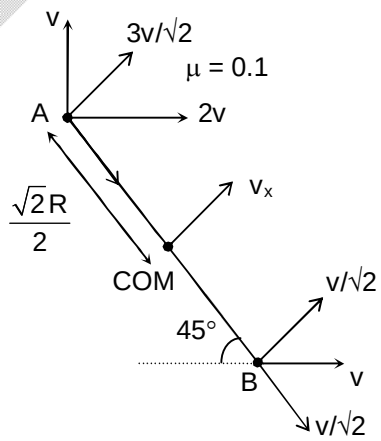
V_x velocity of rod perpendicular to the rod

$\ell = R\sqrt{2}$

$$K.E_{rod} = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}mv_{cm}^2$$

$$= \frac{1}{2}\frac{m\ell^2}{12}\frac{v^2}{R^2} + \frac{1}{2}m\left[(\sqrt{2}v)^2 + \frac{v^2}{2}\right]$$

$$= mv^2\left[\frac{1}{12}+1+\frac{1}{4}\right] = \frac{4}{3}mv^2$$



25. 2

Sol. $E \propto \frac{z^2}{n^2}$' Brackett series $\Rightarrow n = \infty$ to $n = 4$

Balmer series $\Rightarrow n = \infty$ to $n = 2$

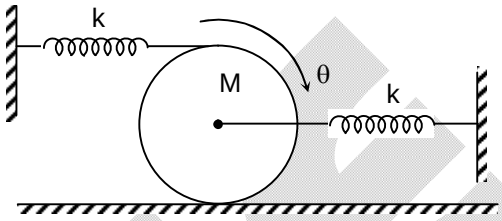
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$h_{f_{breackett}} = \frac{13.6z^2}{16}$ and $h_{f_{Balmer}} = \frac{13.6}{4}$

$\frac{13.6}{4} = \frac{13.6z^2}{15} \Rightarrow z = 2$

26. 3

Sol. $E = \frac{1}{2}k(2R\theta)^2 + \frac{1}{2}k(R\theta)^2 + \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$



$\frac{dE}{dt} = 0$

$\frac{1}{2}k(2)(4R^2\theta)\frac{d\theta}{dt} + \frac{2k}{2}(R^2\theta)\frac{d\theta}{dt} + \frac{2M}{2}v\frac{dv}{dt} + \frac{I}{2}(2\omega)\frac{dw}{dt} = 0$

$\therefore\ 5kR^2\theta\frac{d\theta}{dt} + Mva + I\omega\alpha = 0$

$a = R\alpha$

$\therefore\ 5kR^2\theta(\omega) + M(\omega R)(\alpha R) + I\omega\alpha = 0$

$-5kR^2\theta = (I + MR^2)\alpha$

$\alpha = -\frac{5kR^2}{\frac{MR^2}{2} + MR^2}\theta = -\frac{10k}{3m}\theta$

$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{3m}{10k}}$

27. 9

Sol. $\frac{dA}{dt} \propto r^2$

$\frac{\left(\frac{dA}{dt}\right)_A}{\left(\frac{dA}{dt}\right)_B} = 9$

28. 2

Sol. I maximum when z is minimum at $X_L = X_C$

$I_{rms} = \frac{V}{R},\quad R = 4\Omega$

for DC = $I = \frac{V}{R_{total}};\ I = \frac{12}{4+2}$

$\Rightarrow I = 2$ A

29. 4

Sol. Sound level = $10\log\frac{I_1}{I_0}$

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$70 = 10\log\frac{I_1}{I_0}$

$7 = \log\frac{I_1}{I_0}$

$I_1 = 10^7\, I_0$

Sound level = $10 = 10\ \ell\text{og}\frac{I_2}{I_0}$

$1 = \log\frac{I_2}{I_0}$

$10' I_0 = I_2$

Since $I_2 = I_1\frac{r_1^2}{r_2^2}$

$\therefore\ 10I_0 = I_1\left(\frac{r_1^2}{r_2^2}\right)$

$10I_0 = 10^7 I_0\left(\frac{4^2}{x^2}\right)$

$x^2 = 4^2(10^6)$

$x = 4(10^{-3})\text{m}$

x = 4 km

30. 5

Sol. Let P_1 be the number of harmonics in the first and p_2 be the number of harmonics in the second are

$\therefore\ p_1\sqrt{T_1} = p_2\sqrt{T_2}$

$\sqrt{T_2} = \frac{p_1}{p_2}\sqrt{T_1}$

$\sqrt{Mg} = \frac{5}{3}\sqrt{9g}$

$\therefore\ \sqrt{M} = 5$

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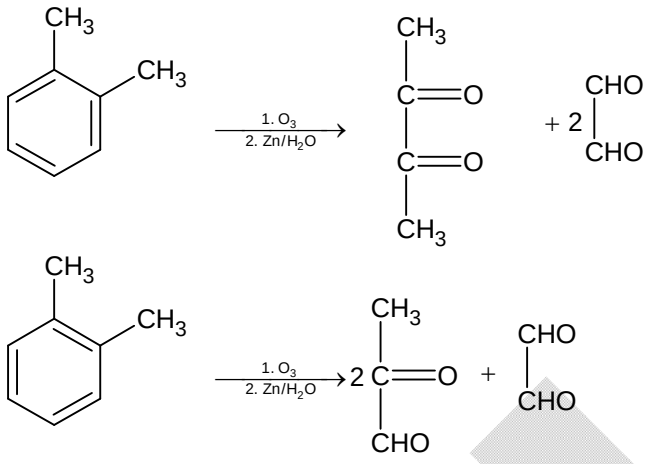
Chemistry

PART – B

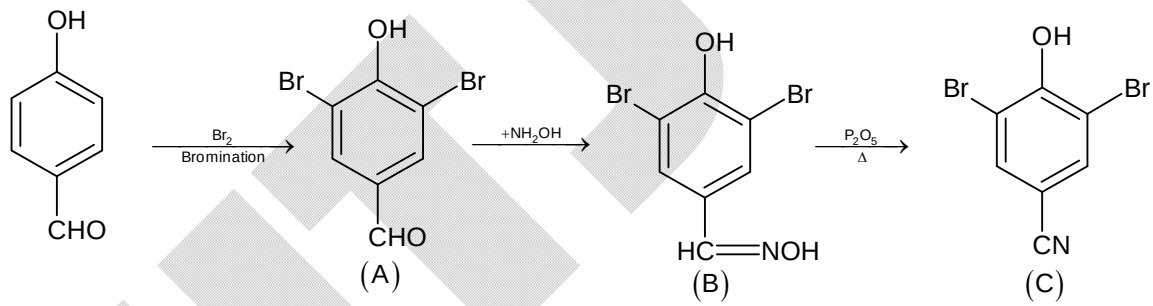
SECTION – A

31. A
Sol. Only in (A) option first complex is trans and second is meridonial.

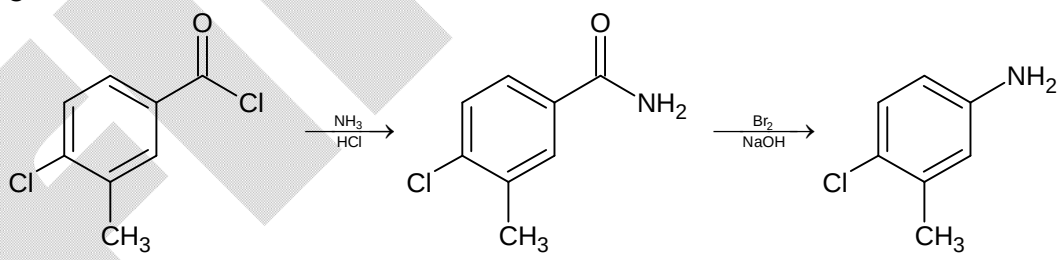
32. C
Sol.



33. B
Sol.

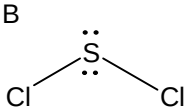


34. C
Sol.



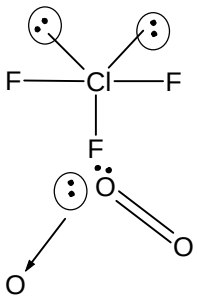
35. A
Sol. $Sc^{+3} \rightarrow 3d^{o}$
$Zn^{+2} \rightarrow 3d^{o}$

36. B
Sol.



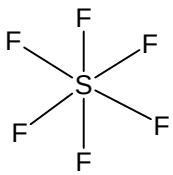
2ℓp electron

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2ℓp electron

1 ℓp electron



No ℓp electron

37. D
Sol. In A, B, C options halide attach with benzene ring.

38. C
Sol. $Na < Al < Mg < Si$

$\longrightarrow$

39. D
Sol. $Na_2S + \left[Na_2Fe(CN)_5NO\right] \longrightarrow Na_4\left[Fe(CN)_5NOS\right]$

Sodium nitroprusside (Violet colour) Sodium thionitroprusside

40. B
Sol. NO, having 7 electrons on nitrogen atom.
H_2SO_4 having expanded octet.

41. C
Sol. Number of angular node $= \ell$
Number of radial node $= n - \ell - 1$

42. D
Sol. Delocalisation require conjugation.

43. C
Sol. Same charge at adjacent element decrease stability.

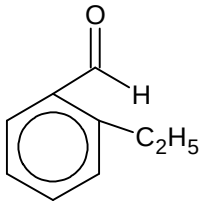
44. D
Sol. 2, 3 are more stable than 1, 4 due to higher aromaticity of ring.

45. D
Sol. d-Orbital not present in carbon.

46. B
Sol. - ve electron gain enthalpy of

Se	S
−195 kJ / mol	−200 kJ / mol
Te	Po
−190 kJ / mol	−174 kJ / mol

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47. C
Sol.  can show all these test.

48. D
Sol. If $\ell = 2, m = -2, -1, 0, +1, +2$ only.

49. D
Sol. Electronegativity not involve in Born-Haber cycle.

50. A
Sol. It is Cross Aldol condensation reaction.

SECTION – B

51. 8
Sol. 2, 3, 4, 6, 7, 8, 9, 11 can show resonance.

52. 3
Sol. $E = \dfrac{30}{\left(\dfrac{2\times 2}{2+2}\right)} = 30$

53. 5
Sol. 1, 5, 6, 7, 9 are aromatic.

54. 9
Sol. FeO, CrO and Fe are not amphoteric.

55. 5
Sol. 1, 2, 5, 6, 9 can show +M effect.

56. 4
Sol. H_2O, NH_3, CH_4, XeF_4 not having π bond.

57. 4
Sol. Iron present in Sodium Nitroprusside, Brown ring complex, Haematite, Magnetite only.

58. 9
Sol. All are correctly matched.

59. 3
Sol. ${}_1H^3$ having 1-proton and 2-neutrons.

60. 2
Sol. Only Be_2C, Al_4C_3 after hydrolysis forms CH_4 gas.

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Mathematics

PART – C

SECTION – A

61. A

Sol. $\because (\vec{a}\times\vec{b})\times\vec{c}$ is a vector which is coplanar with $\vec{a}$ and $\vec{b}$ and perpendicular to $\vec{c}$

Let $\vec{a}=3\hat{i}-2\hat{j}+\hat{k}$, $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$ and $\vec{c}=5\hat{i}+2\hat{j}+4\hat{k}$

Unit vector = $\dfrac{(\vec{a}\times\vec{b})\times\vec{c}}{|(\vec{a}\times b)\times\vec{c}|}$

$(\vec{a}\times\vec{b})=-2\hat{j}-4\hat{k}$

$(\vec{a}\times\vec{b})\times\vec{c}=\hat{i}(-8+8)-\hat{j}(20)+\hat{k}(10)=-20\hat{j}+10\hat{k}$

Now unit vector = $\dfrac{-20\hat{j}+10\hat{k}}{\sqrt{400+100}}=\dfrac{-20\hat{j}+10\hat{k}}{10\sqrt{5}}=\dfrac{-2}{\sqrt{5}}\hat{j}+\dfrac{1}{\sqrt{5}}\hat{k}$

62. D

Sol. $\dfrac{d(x^{3/2})+d(y^{3/2})}{d(x^{3/2})-d(y^{3/2})}=\dfrac{y^{3/2}}{x^{3/2}}$

$u=x^{3/2}$ and $v=y^{3/2}$

$\dfrac{du+dv}{du-dv}=\dfrac{v}{u}$

$\dfrac{udu+vdv}{u^2+v^2}=\dfrac{vdu-udv}{u^2+v^2}$

$\left(\dfrac{d(u^2+v^2)}{u^2+v^2}\right)=-2d\left(\tan^{-1}\left(\dfrac{v}{u}\right)\right)+c$

On integrating, we get

$\ln(u^2+v^2)=-2\tan^{-1}\left(\dfrac{v}{u}\right)+c$

$\dfrac{1}{2}\ln(x^3+y^3)+\tan^{-1}\left(\dfrac{y}{x}\right)^{3/2}=c$

Here $f(x, y)=x^3+y^3$, $g(x, y)=\tan^{-1}\left(\dfrac{y}{x}\right)^{3/2}$.

63. C

Sol. $L=\lim\limits_{x\to0}\dfrac{4\sin^4 x}{x\left(2\tan x-\dfrac{2\tan x}{1-\tan^2 x}\right)}\dfrac{e^{ax}-1}{ax}\cdot a=-2a$

hence $a=-2$

64. A

Sol. $\operatorname{cosec}2^k x=\dfrac{1}{\sin 2^k x}=\dfrac{\sin(2^k x-2^{k-1}x)}{\sin 2^{k-1}x\sin 2^k x}$

$=\cot 2^{k-1}x-\cot 2^k x$

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$f(x)=\sum_{k=1}^{n}\operatorname{cosec}2^k x=\sum_{k=1}^{n}\left(\cot 2^{k-1}x-\cot 2^k x\right)=\cot x-\cot 2^n x$

$g(x) = f(x) + \cot 2^n x = \cot x$

$\lim_{x\to 0^+} H(x)=\lim_{x\to 0}\frac{e^x+e^{-x}-2\cos x}{x\sin x}=2$

$\lim_{x\to 0^-} H(x)=\lim_{x\to 0^-}\left[(\cos x)^{\cot x}+(\sec x)^{\operatorname{cosec} x}\right]$

Let $u(x) = (\cos x)^{\cot x}$

$\lim_{x\to 0^-} u(x)=e^0=1$

similarly $\lim_{x\to 0^-}(\sec x)^{\operatorname{cosec} x}=1$

$\lim_{x\to 0} H(x)=2$

65. B

Sol. $\Rightarrow$ $f(x) = \ln\left(\frac{2+x}{2-x}\right)$

$f(x) = kf\left(\frac{4x}{2+\frac{x^2}{2}}\right)$

$\ln\left(\frac{2+x}{2-x}\right)=2k\ln\left(\frac{2+x}{2-x}\right)$

$k = \frac{1}{2}$

66. C

Sol. Chord whose midpoint is (2, 2) will be of minimum length

Equation of chord = $T = S_1$

$x + y - 4 = 0$

$\Rightarrow m = -1$

On solving chord with circle

$2x^2 - 8x + 1 = 0$

$|x_1 - x_2|=\frac{\sqrt{64-8}}{2}=\frac{\sqrt{56}}{2}=\sqrt{14}$

Length of chord $= |x_1 - x_2|\sqrt{1+m^2}=\sqrt{14}\sqrt{1+1}=2\sqrt{7}$

67. A

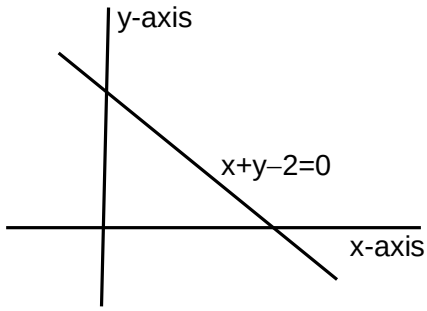
Sol. $xy(x + y) - 2xy$

$xy(x + y - 2) = 0$

$\because$ $\alpha \ge 0$, $\alpha^2 \ge 0$ and $\alpha + \alpha^2 - 2 \le 0$

$\alpha \in [-2, 1]$

Largest value of $\alpha = 1$



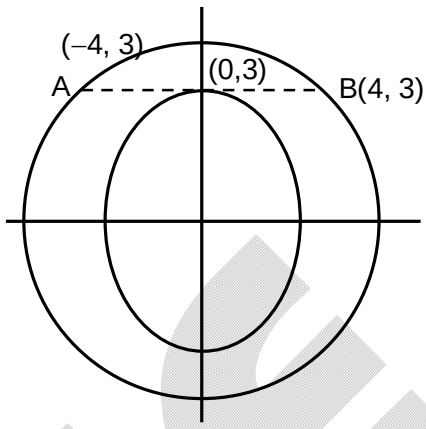
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68. A

Sol. The minimum length of intercept will be possible when y = 3 or = –3 at y = 3, x = ± 4

A(4, 3) and B(–4, 3)

AB = 8



69. D

Sol. Given ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$

$e=\sqrt{1-\frac{3}{4}}=\frac{1}{2}$

focus = (±1, 0)

Length of hyperbola transverse axis = 2a' = 2sinθ $\Rightarrow$ a' = sinθ

a'e' = 1 $\Rightarrow$ sinθe' = 1 $\Rightarrow$ e' = $\frac{1}{\sin\theta}$

$\because$ b'^{2} = a'^{2}(e'^{2} – 1) $\Rightarrow$ b'^{2} = cos^{2}θ

$x^2\text{cosec}^2\theta - y^2\sec^2\theta = 1$

70. C

Sol. P(x) is polynomial of degree 4 having x = 1, 2, 3 as critical points

P'(x) = A(x – 1) (x – 2) (x – 3)

$P(x)=\frac{A}{4}\left(x^4-8x^3+22x^2-24x-55\right)$

$\int_{-2}^{2}P(x)dx=\frac{-1348}{15}\Rightarrow A=4$

$\Rightarrow P(x)=x^4-8x^3+22x^2-24x-55$

tangent at x = –1 is y = –96x – 96

y intercept = –96

71. C

Sol. Favourable cases = 6 = n(A)

Let n(A) = {123, 132, 231, 213, 313, 321}

Samples space = number of ways the ball can be placed in A, B, C = $^{6+3-1}C_{3-1} = {}^8C_2$

$P(A)=\frac{6}{{}^8C_2}=\frac{3}{14}$

72. B

Sol. 9 + 3(sint + cost) + sint cost = 10

3(sint + cost) = 1 – sint cost

9 + 9sin2t = 1 + sin^{2}t cos^{2}t – 2sint cost

$\Rightarrow$ sin^{2}2t – 40sin2t – 32 = 0

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$\sin 2t = \dfrac{40 \pm \sqrt{1600+128}}{2} = 20 \pm \sqrt{432}$

$\sin 2t = 4\left(5 - 3\sqrt{3}\right)$

73. C

Sol. $f(x) = \dfrac{2\sin x}{\sqrt{1-\cos^2 x}} + \dfrac{4\cos x}{\sqrt{1-\sin^2 x}} + \dfrac{8\tan x}{\sqrt{\sec^2 x - 1}} + \dfrac{16\cot x}{\sqrt{\text{cosec}^2 x - 1}}$

$$f(x) = \begin{cases} 30 & , \quad x \in I^{st} \text{ quad.} \\ -26 & , \quad x \in II^{nd} \text{ quad.} \\ 18 & , \quad x \in III^{rd} \text{ quad.} \\ -22 & , \quad x \in IV^{th} \text{ quad.} \end{cases}$$

74. A

Sol. $\because$ BC = $I_{3\times 3}$ and TrA = 27

$S = \text{TrA}\left(1 + \dfrac{3}{4} + \dfrac{6}{4^2} + \dfrac{10}{4^3} + \ldots\ldots\right)$

$S = 27 \cdot \dfrac{64}{27} = 64$

75. C

Sol. $\text{Var}(x) = \dfrac{\sum_{i=1}^{15}(x_i - \lambda)^2}{n} - \left(\dfrac{\sum(x_i - \lambda)}{n}\right)^2$

$\text{Var}(x) = \dfrac{116}{225}$

$\text{S.D.} = \dfrac{2\sqrt{29}}{15}$

76. C

Sol. By Lagrange's Mean Value Theorem $\exists$ c $\in$ (0, 1)

such that $f'(c) = \dfrac{f(1) - f(0)}{1-0}$

$e^{c^2} = f(1) - 4$

$f(1) = e^{c^2} + 4 > 5$

For c $\in$ (0, 1), $e^{2c^2} < e$

$f(1) < 4 + e$

$5 < f(1) < 4 + e$

77. B

Sol. $f(2x + 2y)\sin(x - y) = f(2x - 2y)\sin(x + y)$

$2(x + y) = u$, $2(x - y) = v$

$\dfrac{f(u)}{\sin\frac{u}{2}} = \dfrac{f(v)}{\sin\frac{v}{2}} = k$ for all u, v $\in$ R

$\Rightarrow f(u) = k\sin\dfrac{u}{2}$

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$f'(u) = \frac{k}{2}\cos\frac{u}{2}$

$f''(u) = -\frac{k}{4}\sin\frac{u}{2}$

$4f''(x) + f(x) = 0$

78. D

Sol. Apply $C_1 + C_2$

then $R_2 - R_1$ and $R_3 - R_1$

we get $\begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$

$\Rightarrow 4\sin4\theta + 2 = 0$

$\sin4\theta = -\frac{1}{2}$

$\theta = \frac{7\pi}{24}, \frac{11\pi}{24}$

79. A

Sol. $\Rightarrow \frac{1}{2}\sum_{n=3}^{20} {}^{n}C_3\left(\sin^{2n}x + \text{cosec}^{2n}x\right)$ $\because$ $\sin^{2n}x + \text{cosec}^{2n}x \geq 2$

$\geq {}^{3}C_3 + {}^{4}C_3 + {}^{5}C_3 + \ldots\ldots + {}^{20}C_3 = {}^{21}C_4$

80. 8

Sol. $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 5 \\ 1 & -1 & 1 \end{vmatrix} \Rightarrow \vec{n} = 6\hat{i} + 3\hat{j} - 3\hat{k}$

so equation of plane $6x + 3y - 3z = d$

It passes through (0, 2, 1)

$d = 3$

$\Rightarrow 6x + 3y - 3z = 3 \Rightarrow 2x + y - z = 1$

Given plane $Ax + y - z = d$

$\Rightarrow A = 2$ $\because$ both are parallel

distance between them $\sqrt{6} = \frac{|d-1|}{\sqrt{4+1+1}}$

$\Rightarrow |d - 1| = 6$

so $|d - 1| + A = 6 + 2 = 8$

SECTION – B

81. 7

Sol. $I = \int_0^{\infty} \frac{2x^2e^x}{e^{2x}-1}dx = 2\int_0^{\infty} \frac{(e^x+1-1)x^2}{(e^x+1)(e^x-1)}dx = 2\int_0^{\infty} \frac{x^2}{e^x-1}dx - \int_0^{\infty} \frac{(x/2)^2}{e^x-1}dx$

$I = \left(2 - \frac{1}{4}\right)J$ and $\frac{4I}{J} = 7$

82. 1

Sol. $t_1 + t_5 = t_2 + t_4 = 2t_3$

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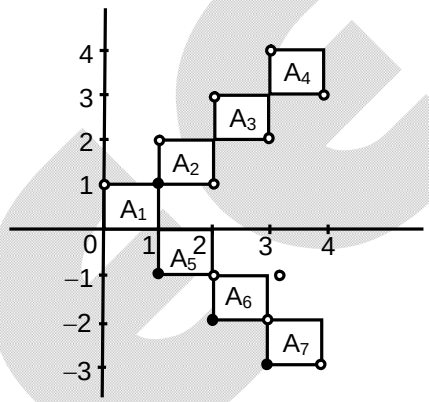
$$\frac{4(t_1 - t_2 - t_4) + 6t_3 + t_5}{3t_1}$$

$$= \frac{4(t_1 - t_1 - t_5) + 3(t_1 + t_5) + t_5}{3t_1}$$

$$= \frac{-4t_5 + 3t_1 + 3t_5 + t_5}{3t_1} = 1$$

83. 7

Sol. Each square is of 1 unit area. So total area = 7



84. 37

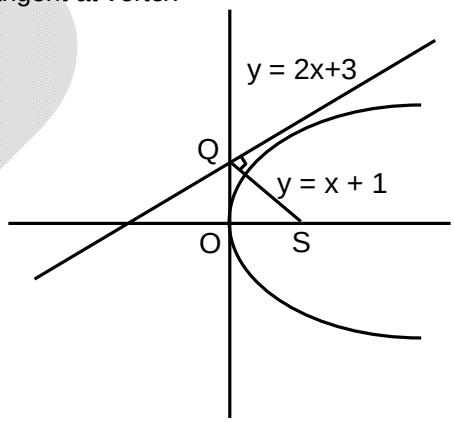
Sol. ∵ foot of perpendicular from focus to tangent lies on tangent at vertex

$$S = \left(\frac{17}{9}, \frac{26}{9}\right)$$

$$OS = \sqrt{\left(\frac{3}{2} - \frac{17}{9}\right)^2 + \left(\frac{5}{2} - \frac{26}{9}\right)^2} = \frac{7\sqrt{2}}{18}$$

$$1LR = 4 \times \frac{7}{18}\sqrt{2} = \frac{14\sqrt{2}}{9}$$

$$\left(\frac{x - y + 1}{\sqrt{2}}\right)^2 = LLR\left(\frac{x + y - 4}{\sqrt{2}}\right)$$



$$\frac{b}{a} = \frac{14 \times 2}{9} = \frac{28}{9}$$

a + b = 37

85. 2

Sol. f(x) = cos2x

$g(\sin 2t) = \sqrt{1 + \sin 2t}$

$g(x) = \sqrt{1 + x}$, $-1 \le x \le 1$

$g(f(x)) = \sqrt{1 + \cos 2x} = \sqrt{2}|\cos x|$

so range of g(f(x)) is $[0, \sqrt{2}]$

$a^2 + b^2 = 2$

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86. 23

Sol. AM ≥ GM

$\Rightarrow$ $x^2y^3z^5 \le 2^2 3^3 5^5$

$N = 2^2 3^3 5^5$

$\because$ $(4n + 2) = 2(2n + 1), n \in N$

$= 1(3 + 1)(5 + 1) = 24$

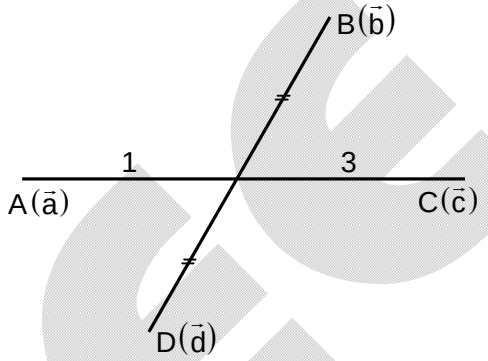
Number of divisors of form $4n + 2, n \in N = 24 - 1 = 23$

87. 2

Sol. $3\vec{a} + \vec{c} = 2\vec{b} + 2\vec{d}$

$\frac{3\vec{a} + \vec{c}}{4} = \frac{\vec{b} + \vec{d}}{2}$

$x + y = 1 + 1 = 2$



88. 3

Sol. $2\sin^2\theta - \cos 2\theta = 0$ and $\sqrt{3}\sin 2\theta = 2\cos^2\theta$

$\frac{1}{2} = \cos 2\theta$ and $\sqrt{3}\sin 2\theta = \cos 2\theta + 1$

$\cos 2\theta = \cos\frac{\pi}{3}$ and $\cos\left(2\theta + \frac{\pi}{3}\right) = \cos\frac{2\pi}{3}$

Hence the number of common solution is 3 which are $\theta = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}$

89. 2

Sol. $(1 + z)^n = z^n$

$|1 + z|^n = |z|^n$

$|1 + z| = |z|$

$|z + 1| = |z|$

so roots are perpendicular bisector of line joining (0, 0) and (0, –1). So roots lie at $x = -\frac{1}{2}$ line

Hence least distance between curve C and circle is $\frac{1}{2} + \frac{3}{2} = 2$.

90. 3

Sol. $\because$ $\sin\frac{\pi}{18} = \sin 10°$

$\sin 30° = 3\sin 10° - 4\sin^3 10°$

$\Rightarrow 8\sin^3 10° - 6\sin 10° + 1 = 0$

$f(\sin 10°) = 0$

$\Rightarrow a = 8, b = 0, c = -6, d = 1$

$f(1) = a + b + c + d = 3$

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